FYP 2 Roadmap and Gantt Chart

Cost-Efficient Optimization of Open-Source LLM Agents for System-Level Tasks

This roadmap covers an eight-week plan for a three-member team from 5 October through 27 November 2026. The goal is to build a demonstrable prototype for three task agents and a shared Ollama benchmarking workflow. Reaching 60% working is a target, not a guarantee; actual progress depends on integration, hardware, model speed, and data quality.

Duration	Team	Target
5 Oct-27 Nov 2026 (8 weeks)	3 members	Demonstrable 60% working prototype target

Scope and Team Ownership

Each member is the primary owner of one task module. Benchmarking and evaluation are shared responsibilities: Member 1 leads the pipeline, while Members 2 and 3 provide module test cases, logs, and results.

Owner	Primary module	Core responsibility
Member 1	Semantic Scholar AI Research Assistant	Paper search/summarization workflow, citations, research quality evaluation; shared benchmark pipeline lead
Member 2	AI Inventory Management Agent for E-commerce	Inventory data workflow, stock alerts/reorder suggestions, task-specific quality checks
Member 3	Robot Planning Agent with 2D Simulator and A* Path Planning	2D grid simulator, A* planner, LLM task interpretation/planning interface, path validity checks
All members	Shared Ollama benchmark	Five-model selection, common baseline/optimized runs, metrics, integration and demo

Weekly Milestones

Week / dates	Work and owners	Dependencies	Milestone and deliverables
1 | 5-9 Oct	All: scope, use cases, acceptance tests. M1: benchmark design. M2: inventory data/schema. M3: simulator and path-planning requirements.	None	Scope freeze; repo/environment plan; 3-5 measurable scenarios per module; risk and hardware inventory
2 | 12-16 Oct	All: choose five suitable Ollama models against local hardware, licensing/access and task fit. M1: common harness skeleton. M2/M3: module interface and sample data/world.	Week 1 scope	Model shortlist of five with rationale; shared JSON input/output contract; test datasets / maps
3 | 19-23 Oct	M1: search + answer/citation baseline. M2: inventory read + alert/recommendation baseline. M3: 2D grid + A* baseline and LLM instruction parsing. All: wire Ollama and run smoke tests.	Week 2 interfaces and data	One basic end-to-end baseline path per module; first run logs; known-issues list
4 | 26-30 Oct	All: run baseline cases across selected models where hardware permits. M1 leads harness, repeatability and metric capture. Each module owner records failure cases.	Week 3 runnable flows	Baseline dataset/results; latency, token use, task success and module-quality fields recorded
5 | 2-6 Nov	M1: prompt/context reduction experiments. M2/M3: optimize prompts and remove avoidable LLM calls; retain deterministic logic for suitable steps. All: rerun matched cases.	Baseline logs and fixed test cases	Optimized variants; baseline-vs-optimized comparison table; no unsupported quality claims
6 | 9-13 Nov	M1: benchmark runner/report. M2: inventory edge cases. M3: obstacle/no-path and valid-path checks. All: test model/module compatibility, capture errors and fallback behavior.	Week 5 optimized variants	Common pipeline usable by all modules; repeatable comparison runs; prioritized bug list
7 | 16-20 Nov	All: integrate three modules into one demo entry point or clearly linked demos. Improve reliability, UI/CLI clarity and reproducibility.	Week 6 stable module branches	Integrated demo candidate; test evidence; screenshots/logs; demo script and limitations
8 | 23-27 Nov	All: freeze a demonstrable build, run acceptance checklist, prepare report/slides/demo. M1 consolidates metrics; module owners validate their flows.	Week 7 integration	27 Nov checkpoint: prototype, benchmark summary, run instructions, known limitations and next-phase backlog

Gantt Chart

The week-by-week chart below is an editable Word table. Filled cells show the planned work period. An editable Mermaid source follows for use in a Mermaid-compatible editor.

Workstream	Owner	W1 5-9 Oct	W2 12-16 Oct	W3 19-23 Oct	W4 26-30 Oct	W5 2-6 Nov	W6 9-13 Nov	W7 16-20 Nov	W8 23-27 Nov
Scope, use cases, and acceptance tests	All								
Select five suitable Ollama models	All								
Module interfaces and fixed test sets	All								
Common benchmark pipeline	M1 lead; all contribute								
Research search and answer baseline	Member 1								
Research citation and quality evaluation	Member 1								
Inventory data workflow and baseline	Member 2								
Inventory alerts and recommendation checks	Member 2								
2D simulator and A* baseline	Member 3								
LLM task parsing and path validation	Member 3								
Prompt/context optimization and matched reruns	All								
Reduce unnecessary calls and test edge cases	All								
Integration, validation, and demo freeze	All								
Acceptance review, report, and final demo	All								

Legend: All / shared Member 1 Member 2 Member 3 Integration / final

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gantt
title FYP 2 Roadmap | 5 Oct to 27 Nov 2026
dateFormat YYYY-MM-DD
axisFormat %d %b
section Shared foundation
Scope and acceptance tests :a1, 2026-10-05, 5d
Select five Ollama models :a2, after a1, 5d
Common benchmark pipeline :a3, 2026-10-12, 25d
Baseline experiments :a4, 2026-10-19, 12d
Optimization and matched reruns :a5, 2026-11-02, 12d
Integration, validation and demo :a6, 2026-11-16, 12d
section Member 1 | Research assistant
Search and answer baseline :b1, 2026-10-12, 2026-10-23
Citation and quality evaluation :b2, 2026-10-26, 2026-11-06
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Optimize prompts and context :b3, 2026-11-02, 12d
section Member 2 | Inventory agent
Data workflow and baseline :c1, 2026-10-12, 2026-10-23
Alerts and recommendation checks :c2, 2026-10-26, 2026-11-06
Reduce calls and handle edge cases :c3, 2026-11-02, 12d
section Member 3 | Robot planning
2D simulator and A* baseline :d1, 2026-10-12, 2026-10-23
LLM task parsing and path checks :d2, 2026-10-26, 2026-11-06
Optimize calls and failure handling :d3, 2026-11-02, 12d
section Final checkpoint
Integrated demo and acceptance review :e1, 2026-11-16, 12d
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Benchmark Design and Metrics

Compare the five models with consistent prompts, task cases, and hardware wherever possible. Record each exact Ollama tag, quantization, hardware, run date, and settings. If time or hardware cannot support every combination, run a small smoke set across all five models first, then repeat the full set for the strongest candidates. State the actual coverage in the report.

Metric	Measurement approach	Module-specific quality examples
Latency	Request-to-final-output wall-clock time; report median and p95 when there are enough repeated runs	All: usable response time
Token usage	Prompt plus generated token counts, or a consistent proxy if the runtime does not expose exact counts	All: tokens per successful task
Task success	Pass/fail against predefined acceptance checks; report success rate	Research: answer grounded in retrieved papers; Inventory: valid action/recommendation; Robot: valid route or correct no-path result
Task-specific quality	Rubric score with evidence; human review for a small fixed sample	Research: relevance/citation correctness; Inventory: stock logic and recommendation correctness; Robot: path validity, goal reached, collision-free route
Cost-efficiency	Compare quality and success with latency/token use; show trade-offs rather than a single opaque score	Baseline vs optimized for each model/module

Optimization experiments should shorten prompts or context, reduce unnecessary LLM calls through skipping, caching, or deterministic rules, and call a model only when the task needs it. Compare every optimization with its baseline on the same test cases. If success or quality declines, do not describe it as an improvement.

Dependencies and Working Rhythm

Dependency	Why it matters	Owner/action
Ollama installed; model storage and RAM/VRAM available	Model downloads and inference speed affect the schedule	All: Week 1 hardware check; M1 logs model tags and runtime settings
Stable module contracts and fixed test cases	Required for matched experiments and integration	All: freeze in Week 2; each module owner maintains test cases
Retrieval/source access and sample research corpus	The research assistant needs reproducible evidence	M1: validate accessible metadata/source strategy and citations
Inventory sample data; robot maps/scenarios	Task-specific checks depend on these inputs	M2: versioned sample inventory; M3: versioned grid maps and expected paths
Regular integration point	Reduces the risk of a last-week merge	All: hold a shared demo/review weekly; begin short daily integration checks in Week 6

What 60% Working Means by 27 November

Treat 60% as a practical scope-completion checkpoint, not product readiness or a guaranteed grade. The target is met when the following observable checks are mostly passing:

- Each module runs at least one end-to-end happy-path demo: a research query returns an answer with citations; inventory input produces a stock/reorder suggestion; a natural-language robot task produces an A* route in the 2D simulator.
- Each module has predefined representative test cases and pass/fail evidence, with basic failures and limitations documented. Aim for at least a 60% pass rate on the fixed set of core happy-path cases.
- Five suitable Ollama models are selected and documented. The shared benchmark pipeline records baseline and optimized runs. Disclose the actual coverage if all five models cannot be tested fully.
- A baseline-versus-optimized comparison is available for latency, token usage, task success, and task-specific quality; at least one optimization is evaluated in each module.
- Reproducible run instructions, a demo script, sample data/maps, a results summary, and a next-phase backlog are ready.

If model compatibility, retrieval quality, hardware limits, or integration defects prevent these checks from passing, report the measured completion level and blocked items. A small, repeatable demo on 27 November is stronger evidence than three incomplete, polished-looking flows.

Risks and Schedule Controls

Risk	Control
The three-module scope may exceed eight weeks	Build one happy path per module first; leave optional UI, advanced features, and production deployment out of scope.
Five models may be slow or unavailable on local hardware	Check hardware in Weeks 1-2; shortlist five models; run a smoke set on all and full runs on feasible candidates; record limitations.
Optimization may reduce answer quality or success	Use a matched test set, quality rubric, and rollback point; latency/token reduction alone is not success.
Late integration and inconsistent interfaces	Freeze contracts in Week 2, run smoke integration in Week 3, deliver the shared pipeline in Week 6, and freeze the demo in Week 7.
External data/API changes	Use versioned local test fixtures and graceful fallback; do not make external access a critical demo dependency.

Checkpoint Reporting Template

At each weekly review, the three owners should report planned versus completed work, demo evidence, benchmark coverage, current blockers, and one measurable outcome for the next week. Estimate percentages using evidence, not task counts alone.

Week ending	Module status / evidence	Benchmark coverage	Blocker and next outcome
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____